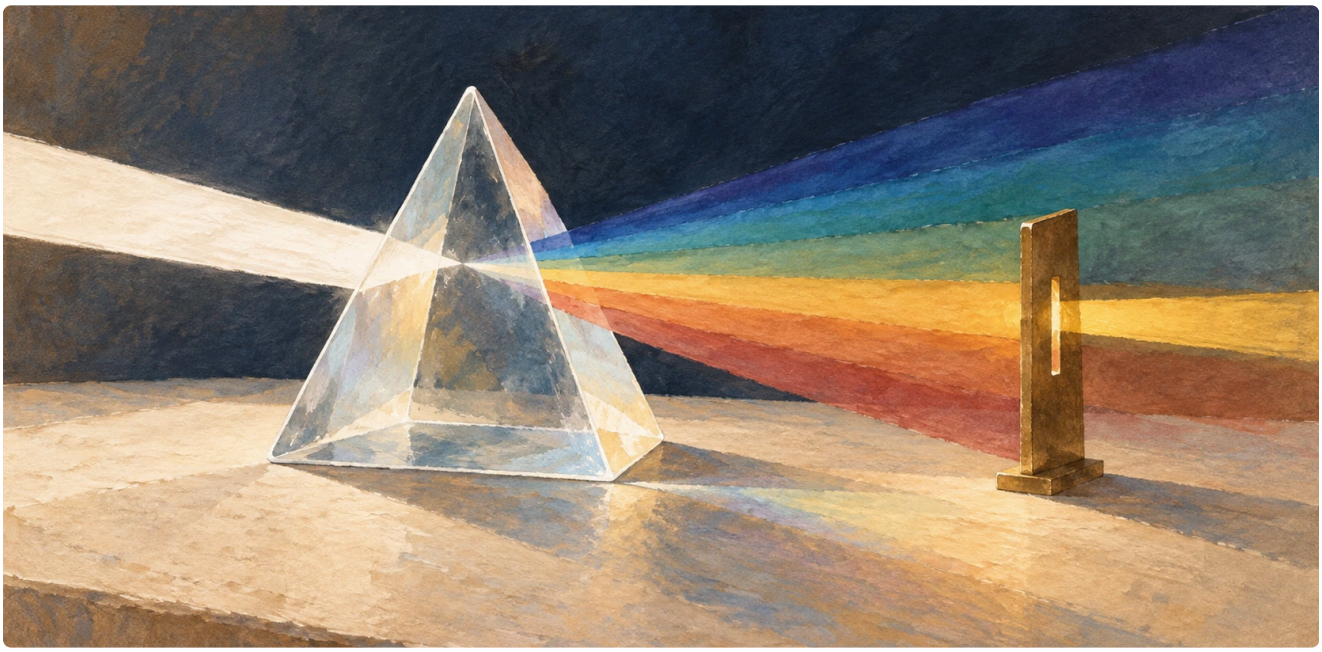
AI Automation and Scientific Judgment in Quantum and Materials Research

AI systems are beginning to turn research questions into calculations and figures, and to predict the electronic structure of materials more quickly. Turning those outputs into knowledge requires joint attention to questions, conditions, learning objectives and verification methods. Results in quantum and materials research, alongside independent reference calculations and sample-based inspection, reveal how automation changes scientific judgment.

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A scene in which differences obscured by a single combined brightness are spread out for closer inspection. This AI-generated conceptual illustration represents the selection of an observable suited to a physical question; it is neither an experimental apparatus nor electronic density-of-states data.

How does a hydrogen molecule's energy change as its two atoms are gradually pulled apart? Calculating the electronic states at different separations produces a curve that shows where the atoms form a stable bond. It is a familiar problem for students of computational chemistry, yet completing it in unfamiliar software requires considerable preparation. One must choose a model of the molecule, select a method for calculating its electronic states, repeat the calculation at each separation and check the results.

El Agente Cuántico, developed by Ignacio Gustin and colleagues, is an AI system that takes requests of this kind, finds software documentation, writes code, and produces calculations and figures. The researchers demonstrated tasks ranging from a hydrogen molecule's energy curve to the dynamics of quantum systems interacting with an environment, many-body states and error-correction simulations. Specialist agents work with individual computational tools, while a central agent coordinates their activities. [1]

This work suggests a way to reduce the burden of learning the syntax of every program a researcher needs. It also raises a further question: once we can perform more calculations, what can we newly establish? In quantum and materials research, AI is advancing along two routes at once. It can carry out computational procedures, or it can approximate the calculations themselves with a learned model. These routes draw on expertise in different places. The first must preserve scientific intent during execution; the second requires a decision about what the AI should be trained to get right.

Agents that turn questions into calculations

Quantum simulation is a way of calculating the states, energies and time evolution of quantum systems such as molecules and materials. As the number of electrons or spins increases, the space needed to describe their states grows rapidly, forcing choices between exact calculations and approximations. Even when practical methods exist, differences in how programs represent a problem and expose their functions make expertise necessary to connect the calculations.

El Agente Cuántico automates this use of scientific software. It consults documentation and examples for tools such as CUDA-Q, QuTiP and TeNPy as needed, then assembles a calculation. The language model writes code, numerical tools execute it, and agents examine the output. The model is therefore doing something different from guessing electronic energies in prose. The demonstrations in this study used classical simulation and numerical computation. Connections to physical quantum hardware were presented as a later development stage. [1]

For the hydrogen molecule, the system used a variational quantum eigensolver (VQE), which adjusts the parameters of a trial state to lower its energy. The publicly released generated code and results table specify a small STO-3G basis model, a classical CPU simulator and 19 interatomic separations from 0.4 to 2.2 Å. A basis is the choice of a finite set of functions used to represent the electrons' spatial states. The comparison was with energies obtained by directly diagonalizing the matrix for the same model. The difference thus measures how closely VQE approaches that reference within the computational model, rather than its discrepancy from an experiment containing all the physical effects of a real hydrogen molecule. [1, 2]

In the upper panel of Fig. 1, the two energy curves nearly coincide at every separation. The lower panel gives the small reported errors their own axis. Ha denotes the hartree, an atomic-scale unit of energy; mHa and μHa are one thousandth and one millionth of a hartree, respectively. The largest difference listed in the released table is 0.001190 mHa, or 1.19 μHa. Preserving the computational conditions and the comparison makes it possible to inspect a specific discrepancy, beyond the impression that a curve looks smooth. We have replotted the authors' released results; we have not run VQE again. [2]

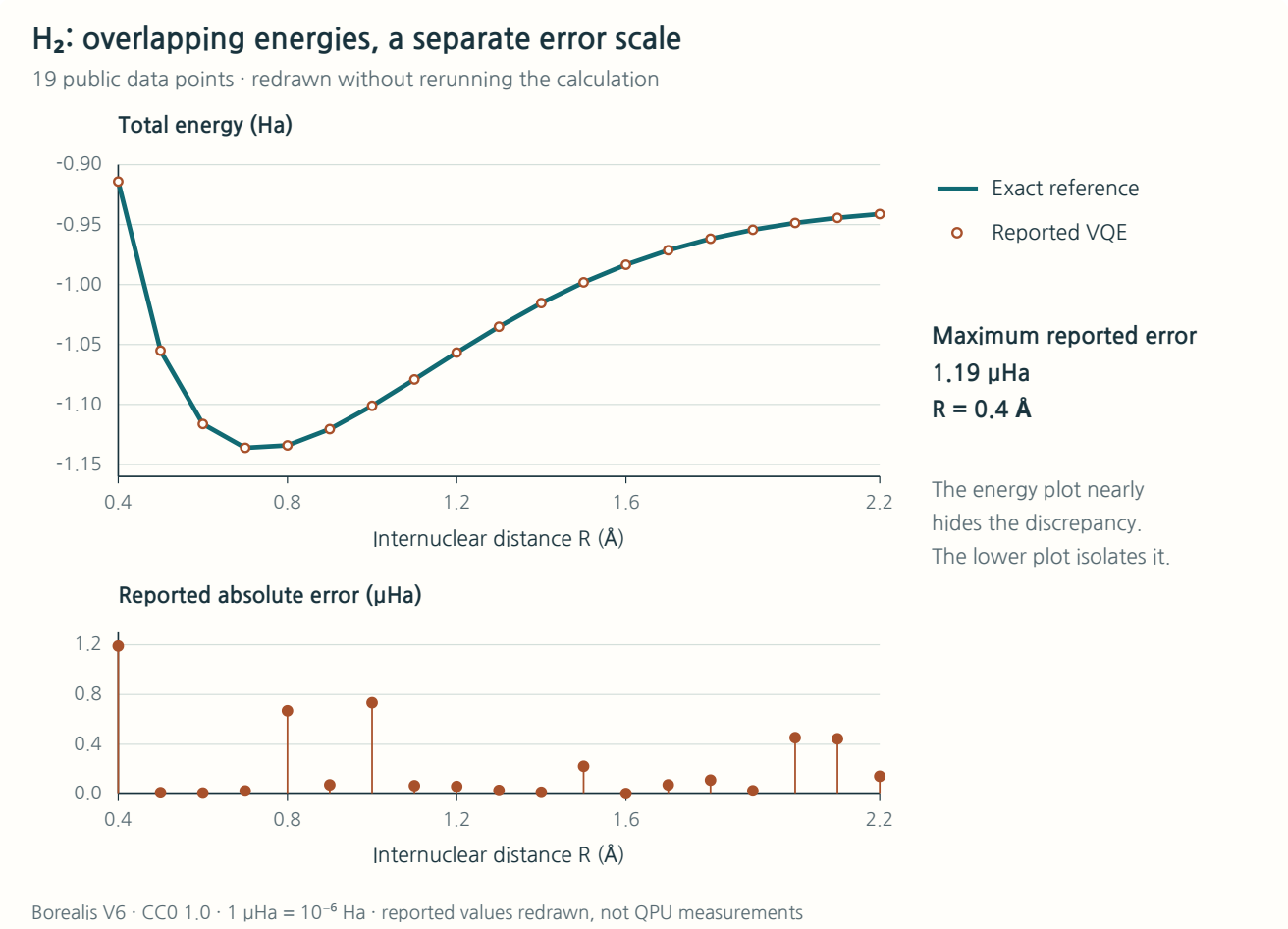


Figure 1. Replot of the 19 results from the hydrogen-molecule demonstration released in Borealis V6. Lines join the points as a visual guide. Energies are compared within the same STO-3G model. The lower-panel errors convert the original table's mHa column to μHa; they are not obtained by subtracting the rounded energy columns. These are reported CPU-simulator results, rather than physical QPU measurements or an independently rerun calculation. [1, 2]

What repeated runs reveal

Starting with a task whose answers can readily be compared, such as a molecular energy curve, helps make both the scope and the limits of automation visible. In a separate benchmark, the researchers ran each of five tasks—VQE, Bell states, Ising dynamics, Lindblad dynamics and hierarchical equations of motion—ten times with identical requests. Runs did not share memory, and one expert per task graded all ten using the same rubric. Partial credit covered implementation, results and validation, and figures and interpretation. The mean scores for the tasks ranged from 97 to 100 out of 100. These scores are not probabilities of task success. [1]

The Lindblad task concerns the evolution of a quantum system under the influence of its surroundings. Quantum oscillations that would persist in an isolated system may decay through environmental interactions, or the system may approach a different steady state. Calculating this behaviour requires specifying both a model of the system and an approximation for the environment's effects. The researchers asked for a comparison of isolated and damped single-qubit evolution, including the z observable as a function of time. This quantity gives the difference in populations of the two basis states.

In one reported run, the system calculated the y component instead of the requested z component. In two other runs of the Ising task, the initial states were random but did not follow the requested Haar distribution. This distribution supplies a way to sample possible pure states without favouring a particular direction. Using random numbers does not by itself produce the same distribution of states. Both examples show why the relationship between a request and its result still needs to be checked after a calculation has run. [1]

An expert cannot judge these cases from the appearance of the plots alone. They must check which component was calculated, how initial states were sampled and whether the environmental approximation suits the research question. Recurring checks can be encoded in software or assigned to another agent. Making clear what those checks must detect is itself a use of expertise.

What should a materials prediction get right?

In materials science, AI can go beyond operating computational software and replace part of an electronic-structure calculation. Kohn-Sham calculations in density functional theory (DFT) repeatedly solve for electronic states and use those states to rebuild the operator used in the calculation. This self-consistent field (SCF) process takes time because the states and operator must be brought into agreement.

A learned Hamiltonian model predicts the matrix of this operator from atomic species and positions. Solving that matrix yields electronic orbitals and energies. This differs from the role of El Agente Cuántico's language-model agents: here, a neural network approximates a numerical result. Realizing the intended benefit requires the predicted matrix to give accurate physical quantities downstream. [3]

FermiScope, developed by Fang Sun and colleagues, revisits the learning objective at this point. A common objective reduces the differences between individual entries in the predicted matrix and their reference values. Yet errors in different entries need not affect the physical quantity of interest equally. In the paper's QHNet diagnostic, additional training to reduce matrix-element error lowered the mean matrix error from 77.56 to 74.44 μHa, while the error in the HOMO-LUMO gap increased from 9,632 to 14,769 μHa. The two errors were measuring different things. [3]

HOMO and LUMO denote the highest occupied and lowest unoccupied orbitals. Their energy difference is important when comparing electronic structures, although it does not always equal a material's optical excitation energy. If the aim is to obtain this orbital gap accurately, the average matrix error alone is an inadequate measure of success.

An electron-density learning objective can also miss a relevant difference. Consider a simple predicted matrix in a fixed two-function basis, with two electrons of opposite spin filling the lower spatial orbital. Keep that orbital's energy at −1 eV and change only the upper orbital's energy from +1 to +2 eV. Because the occupied orbital stays the same, the two predictions have the same electron density, while their orbital gaps change from 2 to 3 eV. **A prediction may appear correct under one comparison while concealing an important difference under another.**

Figure 2 expresses this relationship with two-level matrices in an orthonormal basis. It does not construct two independent self-consistent DFT solutions with the same density. It illustrates how a density comparison can fail to detect a change made only to an unoccupied level in a predicted matrix.

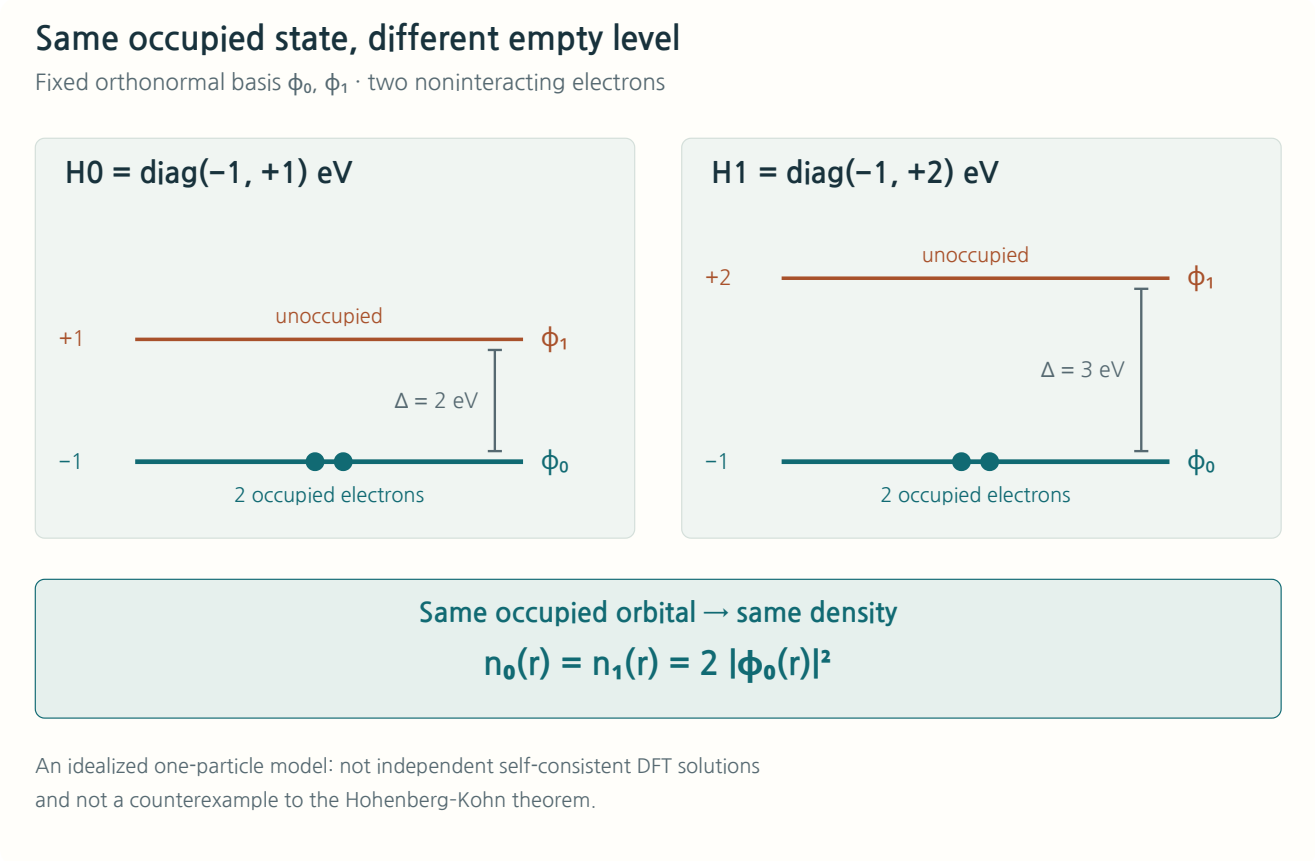


Figure 2. An educational two-level model in the same fixed orthonormal basis. With two electrons of opposite spin filling the lower spatial orbital, unchanged basis functions and unchanged occupation order, the density remains the same while the unoccupied energy and gap differ. This is the review's reconstruction of a limitation of density comparisons discussed by FermiScope. It is neither molecular data nor a pair of independent SCF solutions, and it is not a counterexample to the Hohenberg-Kohn theorem. [3]

FermiScope adds the cumulative local density of states (cumulative LDOS) to the learning objective. At each point in space, this quantity accumulates contributions from orbitals below a chosen energy threshold. The comparison therefore extends from electron density alone to information across energy thresholds. Applying different learning objectives to the same neural-network architecture allows the resulting changes to be tested. [3]

On QH9's standard test set of 13,084 molecules, the three networks' orbital-gap mean absolute errors fell by 49-79% relative to their respective pretrained states. For QHNet, the error decreased from 9,632 to 2,027 μHa; for SPHNet, from 1,554 to 789 μHa; and for WANet, from 1,684 to 791 μHa. Each percentage improvement uses that model's previous error as its denominator. These results do not imply the same improvement for every material or an accuracy of 79%. [3]

The relationship to computational speed needs a separate comparison. In an appendix, the authors report that, for 50 test molecules, using a predicted matrix to initialize SCF reduced the iteration count by about one-third and the total time, including inference, by 13-15% relative to the standard minao initial guess. The pretrained

models and several further-training methods produced similar acceleration. This result therefore cannot establish an additional speedup attributable specifically to cumulative-LDOS training. What the study particularly illustrates is the improvement in a prediction's usefulness that can come from designing a learning objective around a physical quantity. [3]

Preserving conditions that are already correct

Whichever approach is used, certain choices must be preserved when scientific intent is translated into computational conditions. A VASP INCAR file contains choices about magnetism, approximations to electronic interactions, numerical computation and the order of operations. These settings interact, so a file that a program can read does not necessarily describe the intended calculation.

INCARBench, developed by Bin Shao and colleagues, constructed 192 generation cases from 48 materials and four task types, then evaluated 576 repair cases derived from them. These included inputs containing no errors. The aim was to test both the ability to fix incorrect values and the ability to avoid unnecessary changes to settings that already met the criteria. [4]

The benchmark's reference configurations provide standardized evaluation criteria, rather than a unique correct answer among every physically possible choice. The benchmark also does not evaluate convergence or predicted material properties through actual VASP runs. Even so, it makes explicit which conditions should be retained and tests the scope of a repair. Defining what can be automated requires distinguishing a setting adopted for past convenience from a condition needed to preserve the research question. [4]

Reconstructing a calculation from a paper

A different difficulty arises when computational results pass to another researcher. Checking the same claim from a paper may require working out which conditions and procedures were omitted. In AutoMat, Ziyang Huang and colleagues worked with experts to select 85 claims from ten materials-science papers and turn them into reproduction tasks. An agent had to read the paper and supplied materials, prepare an environment, perform the calculations and decide whether the resulting evidence supported the claim. [5]

The highest success rate among five agent configurations was 53%. Success here meant that a separate AI evaluator inspected execution traces and artifacts and awarded at least four out of five points. This cannot be ranked directly against El Agente Cuántico's mean partial-credit scores: the tasks, supplied information, models and tools, and unit of evaluation differ. One study tested repeated execution of specified simulation requests; the other tested the recovery of procedures from papers. [1, 5]

AutoMat also examined its evaluator. Comparing AI scores with experts' blinded assessments of 40 runs gave a weighted kappa of 0.69. This measures agreement for that sample and rubric, rather than guaranteeing agreement with human judgment on every task. The difficulty of recovering executable procedures from paper

text raises a question about where researchers' tacit knowledge remains. Recording expertise as reproducible procedures makes it available to both AI and the next researcher. [5]

What did the AI referees count across 228 papers?

As studies of this kind multiply, Anton Akhmerov's Journal Club commentary, Papers to end all papers, asks how their achievements should be read. Its public methods and data cover 228 relevant arXiv preprints first submitted between 18 April and 18 October 2026 and classified as giving language models a substantive or proposed role. The search looked for AI use in titles and abstracts, so it could miss physics research that disclosed such use only in the body. [6, 7]

Averaging the classifications of two models, about 14% of the papers centred on questions about the physical world. Many of the remaining papers asked about AI capabilities or workflows. Extending this distribution to all science that uses AI would be difficult. The search itself is better at finding studies that put AI in the foreground, and the physical-question category includes mathematical results in quantum information as well as experimental discoveries. [7]

GPT-6.1 Sol referees reading the papers raised at least one major issue in 200 of them, or about 88%. Common concerns included access to materials needed for reproduction, causal claims that did not isolate their proposed cause, and unequal comparison baselines. This is not a human-confirmed error rate. It is a set of model judgments made under a fixed request and rubric. Major issues were also raised in 76% of a control group of 50 ordinary condensed-matter and quantum papers from the same period, compared with 84% of 130 AI papers in the same fields. [7]

Figure 3 needs to be read with both comparison groups and their uncertainty in view. In the public data, the 95% interval for the difference between the two proportions includes zero. Consistency in a model's criticisms is different from agreement with an expert assessment of correctness. A survey like this can help identify questions for closer reading, but it cannot readily serve as a verdict on the validity of individual studies. [7]

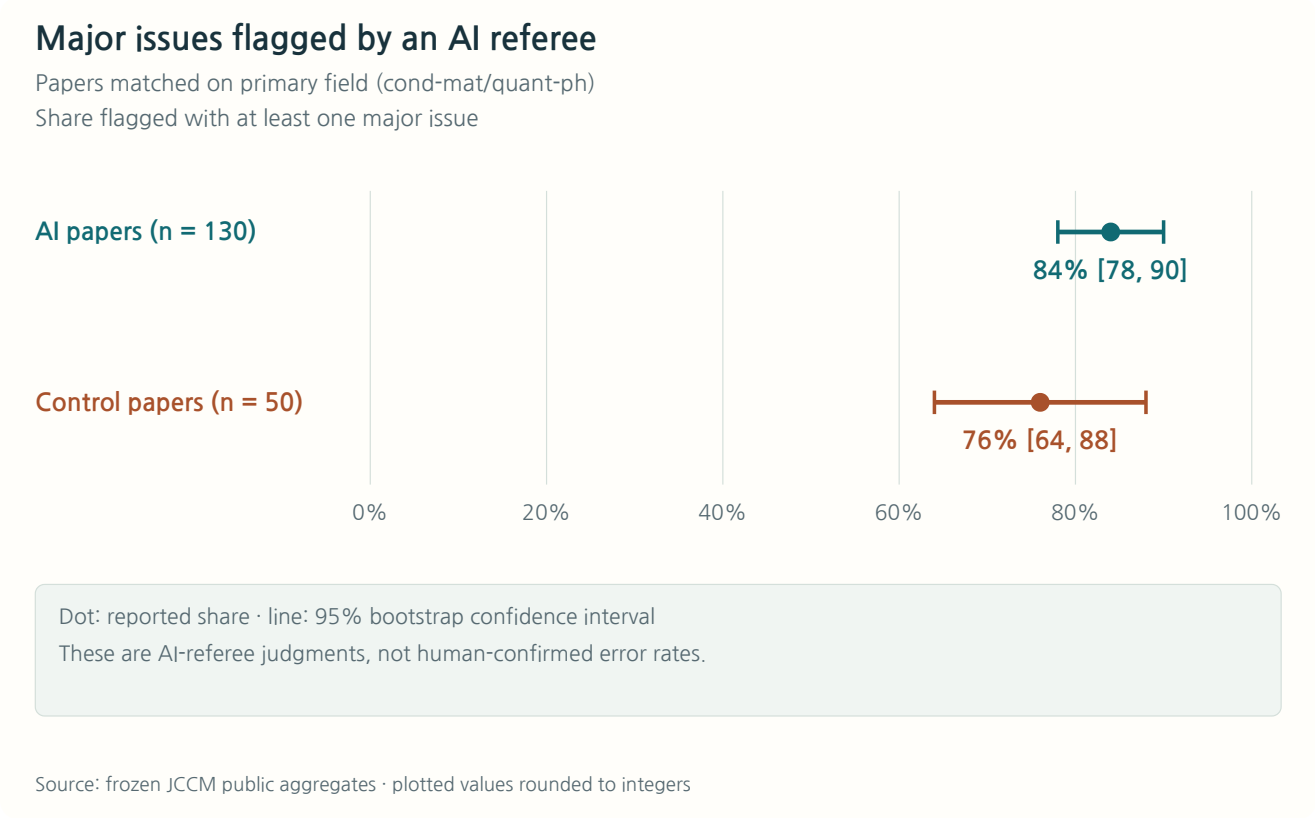


Figure 3. Shares of papers flagged by AI referees for major issues in the survey's same-field samples, with the authors' reported 95% bootstrap intervals. These comparison groups are kept separate from the 88% figure for all 228 papers. The values are not human-confirmed error rates, and the difference between the two proportions cannot establish a causal effect of AI use. The original reviews were not rerun. [7]

Expert judgment also needs comparison and validation

These studies make it possible to describe the expertise required in concrete terms: framing a phenomenon and selecting an observable that distinguishes it, judging when a model's approximations are valid, and designing a comparison that could expose an incorrect result. FermiScope placed this knowledge in a learning objective, INCARBench in criteria for input settings, and AutoMat in reproduction procedures and assessment.

Expert participation does not determine the value of a combination by itself. In a meta-analysis of 106 experiments published in 2020-2023, Michelle Vaccaro and colleagues found that human-AI combinations performed better on average than humans alone, but worse on average than whichever performed better alone, the human or the AI. The study did not directly test today's scientific agents. It nevertheless shows why improving human performance and achieving collaboration that surpasses both components are different questions. [8]

The value added by experts working with AI should likewise be open to comparison in research settings. Under the same time and cost budget, one could compare an expert using ordinary tools, an expert working with AI, and an agent supplied with explicit procedures. An apparent benefit from expert intervention might instead come from better documentation or reference calculations. Counting review, failed runs, retries and corrections alongside coding time would help establish how much capacity automation actually frees.

The evaluation must also be able to expose an expert's mistaken instruction. If an observable does not address the hypothesis, or the supplied conditions are incompatible, testing an agent's ability to carry them out is insufficient. Defining success criteria and tolerances before seeing the results, then comparing with an exact answer for a small system, another implementation or a new measurement, may help. Whether a comparison is independent depends on the assumptions shared in that particular problem. Agreement among several models does not eliminate the possibility of a common error.

How researchers can design verification

If researchers have to repeat every AI-generated calculation from the beginning, reviewing the work could consume all the time saved by automation. Effective verification requires both finding errors in completed results and designing calculations and comparisons that make errors easier to detect. Checking what was calculated, reducing numerical error and establishing whether a model explains a real phenomenon require different kinds of evidence.

In 2026, John D. Jakeman and colleagues proposed recommendations for verification and validation (V&V) in scientific machine learning. The distinction is between checking whether code and numerical methods solve the specified mathematical problem correctly, and assessing whether the model accounts for observations in its intended use. Their article proposes extending established computational-science practices to machine learning, rather than reporting a new experiment on efficiency. Researchers can apply this distinction when reviewing AI-generated results too. [9]

For the tasks in this review, a starting point would be to specify the physical quantity of interest, units, initial conditions and tolerances before examining the results. For a hydrogen-molecule curve, this means identifying which energy is being compared within which basis model. When choosing materials candidates, the relevant question may be how errors in energy differences or orbital gaps affect the choice, rather than the average error across all predictions. If the difference needed to make a decision is comparable to the numerical error, the calculation needs greater precision or the decision should be deferred. Tolerances should not be widened retrospectively to accommodate the AI's output.

Recurring checks can be assigned to software. One example is checking that total probability remains one during a calculation. Yet obtaining a physically possible state does not establish that the requested z observable was calculated. For a verification test, prepare a qubit in the first of its two basis states, $|0\rangle$. At the initial time, the expectation value of the normalized z observable (Pauli Z) is +1, whereas that of the y observable (Pauli Y) is zero. Adding this small test input separately from the original task can distinguish implementations that confuse the two observables. For hydrogen, one can compare against direct diagonalization of the same small Hamiltonian. For a larger calculation, one can vary numerical settings such as the time step, basis or grid, as appropriate, and check **how much the physical quantity of interest changes**. An agent may write these checks, but reference values should be checked against a separate expression, implementation or dataset, rather than obtained by reusing the function under inspection.

Every laboratory need not build all its reference calculations anew. Emanuele Bosoni and colleagues cross-checked 960 equations of state—energy as a function of volume—for 96 elements in ten crystal models per element, using two all-electron codes, then used them to evaluate nine pseudopotential-based approaches. All-electron codes treat core as well as valence electrons, whereas pseudopotentials represent the effect of core electrons through an effective potential. This effort establishes shared workflows and reference data for comparing numerical agreement across implementations. Agreement between implementations of the same DFT approximation, however, does not guarantee agreement with experiment, and a benchmark of equations of state does not automatically extend to other physical quantities. Choosing references appropriate to the quantity and computational conditions remains a task for scientific judgment. [10]

When there are thousands of results, deciding which to inspect becomes another problem. More precise calculations could be prioritized for new elemental combinations, structures outside the training domain or cases in which small errors would reverse the ranking of candidates. At the same time, a random sample should be retained among results that appear unproblematic. Inspecting only suspicious cases can uncover failures there, but makes it difficult to estimate the error rate across all results. There must also be opportunities to discover unanticipated errors among answers the agent considers reliable.

Statistical methods also seek to combine large numbers of predictions with a small set of trustworthy observations. Prediction-powered inference (PPI), developed by Anastasios N. Angelopoulos and colleagues, estimates prediction errors on a subset whose values have been checked through measurements or reference calculations. It combines these errors with the larger set of predictions to construct confidence intervals for quantities such as means and proportions. The original method assumes, among other conditions, independently sampled data from the same population and a predictor independent of those data. Updating a model using the inspection sample and then counting that same sample as an independent check violates this condition. Such an approach could be considered when evaluating large numbers of research outputs or papers, but the original method cannot simply be applied to a selection consisting only of suspicious cases. It neither guarantees the correctness of individual results nor corrects systematic errors in expert judgments themselves. The inspection sample must be representative and its judgment criteria established, with their costs included when assessing the benefit. [11]

Methods for selecting inspection targets more actively are also being studied. In active statistical inference, Tijana Zrnic and Emmanuel J. Candès give greater attention to cases with higher estimated uncertainty, while recording each case's selection probability and using weights to account for unequal sampling. Uniformly random sampling is mixed in so that cases the model considers reliable still have a chance of being inspected. This is a statistical procedure for improving population-level inference from limited real observations. It does not replace direct checks for rare failures that could determine a conclusion. Whether reducing the number of samples that require direct inspection also saves researchers' total review time must be measured separately. [12]

The verification method itself should be open to testing. One could prepare cases in which the requested observable has been changed or a valid setting unnecessarily modified, then measure whether the checks detect them. Recording the rate at which correct results are wrongly rejected, together with review time, would help determine whether adding more checks actually helps. Comparing which errors are detected more

effectively under the same review budget, and which are still missed, is a research problem for scientists and AI developers to pursue together. The laboratory applications proposed in this section build on the cited research; they are not experimental results showing reduced verification costs or errors in these agents.

The expertise passed on to the next researcher

Making these checks part of everyday research requires preserving the reasons for judgments alongside the results. One possible direction is to turn recurring judgments into testable criteria and failure examples, while concentrating review on new assumptions and unexpected results. Researchers need to explain what constitutes an important failure, and system developers need to make execution environments, intermediate results, failures and reasons for changes available for inspection. The amount of output that can be reviewed belongs in the research plan too.

This process also concerns the education of the next generation. If records of correcting AI-generated calculations omit why a result was questioned and what was changed, junior researchers may lose opportunities to learn that judgment. Training can still include predicting a small system's behaviour before comparing it with a calculation, or diagnosing deliberately incorrect initial conditions and observables, even as implementation is automated. Practical knowledge held in an expert's memory can thereby become knowledge that someone else can test.

Lisa Messeri and M. J. Crockett have discussed the conceptual risk that AI could increase scientific output while narrowing understanding. Repeatedly choosing only questions that familiar tools handle well could displace other methods and perspectives. Their work offers a perspective for examining the relationship between productivity and understanding, rather than an experimental result that negates automation's benefits. [13]

Identifying which assets to preserve takes on concrete meaning here. Software interfaces and recurring procedures can change. Accumulated data, models and criteria for recognizing failure can be developed further by recording their conditions and rationale, so that new tools and researchers can use them. Some criteria can become automated checks; some judgments must be learned anew in the face of new evidence. Experts and AI systems alike need to be able to revise familiar answers.

Returning to the hydrogen molecule's energy curve, the researcher has gained both a figure and a remaining question: within which model, at what accuracy and against which reference was this curve obtained? If the resources freed by delegating repeated implementation to AI can be used for better questions and decisive comparisons, automation can broaden the scope of research. Its results accumulate as knowledge when the next researcher can check the same conditions and move on to different ones.

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How this review was made

Responsible editor	Hyun-Jung Kim
AI system	OpenAI Codex; exact authoring-model identifier not retained. Models used in the source research are distinct from this writing system.
Verifiable agent roles	root: Korean writing, conceptual-cover planning, editorial and publication integration; cuantico_source: El Agente Cuántico, AutoMat, public-data crosschecks and scientific audit; dqb_fermiscope: FermiScope, INCAR and materials-calculation conditions, scientific audit; journal_dataset: JCCM aggregates and controls, human-AI collaboration literature and full English translation; agent_standards: reported-data redrawing, teaching-model diagrams, equation/font typesetting, HTML/PDF production and output checks.
Editorial harness	AI Tech Review Editorial Harness 2026.08; Scientific Explanation and Visual Communication Principles 1.0; APS numbered references.
Verification scope	Claims and conditions were compared against papers, official code/documentation and public data. The technical account of El Agente Cuántico uses the arXiv v3 PDF checked on 2026-11-01, not assumed to be the same version as the metadata, HTML or final journal article. Deviations in the observable and initial-state distribution are paper-reported cases, not independently reproduced using public code. The 19 H_2 points in Borealis V6 and JCCM controls are redrawn reported values; the original calculations and AI reviews were not rerun. The density-level diagram is a fixed-orthonormal-basis, two-electron teaching model, not independent self-consistent DFT solutions or a Hohenberg-Kohn counterexample. No actual QPU, paid API or research VASP calculation was run. Bilingual equations, figures, metadata, HTML/PDF and mobile output were checked. The revision compares SciML V&V recommendations, independent DFT-reference workflows, PPI and active-inference sampling conditions. The separate Pauli test input was checked with two-by-two matrices. Proposed laboratory checks and statistical methods were not implemented or benchmarked for time savings.
Human review record	The requested subject and discussion direction informed the review. No line-by-line human approval or reader-comprehension assessment was performed.
Evidence cutoff	2026-11-01

Reported public values, the review's teaching model and the conceptual cover are distinguished. The AI-generated cover is a conceptual illustration, not an actual device or computational dataset.